Supervised learning

Supervised machine learning uses labeled examples. A model learns relationships between input features and target values, then predicts outcomes for previously unseen examples.

Model evaluation

A machine learning model should be evaluated on data that was not used for training. Precision, recall, and F1 score measure different aspects of classification performance.

Overfitting

Overfitting occurs when a model learns details specific to its training examples and performs poorly on new data. Regularization and cross-validation can help identify or reduce this problem.